Dimensions of weight-n newforms for $\Gamma_1(64)$: a proof of Luschny's conjecture on OEIS A063337

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Abstract

OEIS A063337 tabulates the dimension of the space of weight-n cuspidal newforms for $\Gamma_1(64)$. In March 2012 P. Luschny conjectured the closed form $a(n) = 72n - 155$ for $n > 2$. We prove it. The route is the classical dimension formula for $S_k(\Gamma_1(M))$ together with the Atkin–Lehner–Li newform sieve; for $N = 64$ the sieve leaves only a short explicit combination, and the result is $\dim S_k(\Gamma_1(64))^{\mathrm{new}} = 72k - 83$ for $k \geq 2$. As at level 32, there is no parity term, because the level $M = 4$ — the only one carrying a half-integer correction — does not survive the sieve at $N = 64$. We also record that the entry's data is offset by one against its own name: the tabulated row begins at weight 1, whose value is the sentinel -1 because weight 1 admits no dimension formula, so that $a(n)$ is the weight-$(n-1)$ dimension. Under that reading — the only one consistent with the data — the conjecture is exactly right.

1 The sequence and the conjecture

OEIS A063337 (N. J. A. Sloane, 14 July 2001) has offset 2 and begins

$$-1,\ 61,\ 133,\ 205,\ 277,\ 349,\ 421,\ 493,\ 565,\ \ldots$$

The entry carries

> **Conjecture:** $a(n) = 72n - 155$ for $n > 2$. — Peter Luschny, Mar 2012

As of the "Last modified" line on the live entry (23 August 2026) this is still recorded as an unproven conjecture, and no proof appears in the entry's links.

2 The dimension formulas

All of the following is classical; we quote it in the form given by Stein [1], Chapter 6, where it is assembled from Shimura [3, §2.6], Diamond–Shurman [2, Ch. 3] and Miyake [4, §2.5], and derived by Riemann–Roch applied to $X_1(N) \to X_1(1)$.

For a positive integer N put

$$\mu_0(N) = \prod_{p\mid N}\left(p^{v_p(N)} + p^{v_p(N)-1}\right), \qquad \mu_1(N) = \begin{cases} \mu_0(N) & N = 1, 2, \\ \frac{1}{2}\varphi(N)\mu_0(N) & N \geq 3, \end{cases}$$

$$c_1(N) = \begin{cases} c_0(N) & N = 1, 2, \\ 3 & N = 4, \\ \sum_{d\mid N} \frac{1}{2}\varphi(d)\varphi(N/d) & \text{otherwise,} \end{cases}$$

$$g_1(N) = 1 + \tfrac{1}{12}\mu_1(N) - \tfrac{1}{4}\mu_{1,2}(N) - \tfrac{1}{3}\mu_{1,3}(N) - \tfrac{1}{2}c_1(N),$$

where $\mu_{1,2}(N) = \mu_{1,3}(N) = 0$ for $N \geq 4$. Here $g_1(N)$ is the genus of $X_1(N)$ and $c_1(N)$ is its number of cusps.

Proposition 1 ([1, Prop. 6.6]). $\dim S_2(\Gamma_1(N)) = g_1(N)$. *For* $k \geq 3$ *and* $N \geq 3$, *writing* $a(N,k) = (k-1)\big(g_1(N)-1\big) + \big(\frac{k}{2}-1\big)c_1(N)$,

$$\dim S_k(\Gamma_1(N)) = \begin{cases} a(N,k) + \frac{1}{2} & N = 4 \text{ and } k \text{ odd,} \\ a(N,k) + \lfloor k/3 \rfloor & N = 3, \\ a(N,k) & \text{otherwise.} \end{cases}$$

Proposition 2 ([1, Prop. 6.6], Atkin–Lehner–Li).

$$\dim S_k(\Gamma_1(N))^{\text{new}} = \sum_{M \mid N} \bar{\mu}(N/M) \dim S_k(\Gamma_1(M)),$$

where $\bar{\mu}(R) = 0$ *if* $p^3 \mid R$ *for some prime* p, *and otherwise* $\bar{\mu}(R) = \prod_{p \,\|\, R}(-2)$, *the product being over primes exactly dividing* R.

Note that $\bar{\mu}(1) = 1$, $\bar{\mu}(p) = -2$, $\bar{\mu}(p^2) = 1$ and $\bar{\mu}(R) = 0$ once any prime occurs to the third power.

3 The indexing of the entry

The data of this entry comes from Stein's table *Dimensions of the spaces* $S_k^{\text{new}}(\Gamma_1(N))$, linked from the entry. Two features of the data fix how it is to be read.

First, the initial term is -1. A dimension cannot be negative, so this is not a dimension; it is the sentinel for an uncomputed value. Weight 1 is exactly the case for which no dimension formula is available — Stein opens Chapter 6 by saying that he gives no simple formulas for weight 1, and that it may not be possible to give any, since the methods behind all the formulas above fail at $k = 1$.

Second, comparing the remaining terms against Proposition 2 shows that they are the dimensions in weights $2, 3, 4, \ldots$. Hence the tabulated row runs from weight 1, with weight 1 recorded as -1, while the entry's offset is 2. The consequence is

$$a(n) = \dim S_{n-1}(\Gamma_1(N))^{\text{new}} \qquad (n \geq 3), \tag{1}$$

that is, the entry's name is off by one against its own data. This is verified below against every published term, and against the two neighbouring entries drawn from the same table, which exhibit the same shift. We take (1) as the reading of the entry; it is the only one consistent with the data, and it is the reading under which the conjecture is a true statement.

4 The computation at level 64

Lemma 3. *Since* $64 = 2^6$, *the divisors* $M \mid 64$ *with* $\bar{\mu}(64/M) \neq 0$ *are exactly* $M = 64, 32, 16$, *with* $\bar{\mu}(1) = 1$, $\bar{\mu}(2) = -2$, $\bar{\mu}(4) = 1$. *Hence*

$$\dim S_k(\Gamma_1(64))^{\text{new}} = \dim S_k(\Gamma_1(64)) - 2\dim S_k(\Gamma_1(32)) + \dim S_k(\Gamma_1(16)).$$

The relevant invariants are $g_1(64) = 93$, $c_1(64) = 72$; $g_1(32) = 17$, $c_1(32) = 32$; $g_1(16) = 2$, $c_1(16) = 14$. None of the three levels is 3 or 4, so no exceptional term occurs.

Theorem 4. *For every $k \geq 2$,*

$$\dim S_k(\Gamma_1(64))^{\mathrm{new}} = 72k - 83.$$

Proof. Substitute the values of g_1 and c_1 from Lemma 3 into Proposition 1 and form the combination of Proposition 2. Each $\dim S_k(\Gamma_1(M))$ occurring is, for $k \geq 3$, the affine function $(k-1)(g_1(M)-1) + (\frac{k}{2}-1)c_1(M)$ of k, with the single exception noted in Lemma 3; the combination of affine functions is affine, and the exceptional term contributes the stated correction. The case $k = 2$ is checked directly from $\dim S_2(\Gamma_1(M)) = g_1(M)$. □

Corollary 5 (Luschny's conjecture)**.** *For every $n > 2$, $a(n) = 72n - 155$, which is Luschny's conjecture. Indeed $a(n) = \dim S_{n-1}(\Gamma_1(64))^{\mathrm{new}} = 72(n-1) - 83 = 72n - 155$.*

Proof. Combine Theorem 4 with the reading (1) of Section 3. □

Remark (why $n = 2$ is excluded, and correctly so)**.** Under (1), $a(2)$ is the weight-1 value, which is the sentinel -1 rather than a dimension. Luschny's restriction to $n > 2$ is therefore exactly right, and could not be removed: the conjectured formula gives a different value at $n = 2$.

Remark (honest assessment)**.** No new mathematics is involved: the dimension formulas are classical and the newform sieve is Atkin–Lehner–Li. What was missing was that nobody had carried out the substitution. The one genuinely delicate point is the indexing of Section 3; a proof addressed to the entry's name rather than its data would conclude that the conjecture is false, which is why that section is stated before any computation. The natural destination is a comment on the entry, together with a correction of the name or the offset.

5 Verification

A verification script, written separately from this note, checks every claim and prints every number quoted. It (i) implements Propositions 1 and 2 and validates them against Stein's own published tables — eight tabulated values of $\dim S_k(\Gamma_0(N))$, ten of $\dim S_k(\Gamma_1(N))$, and the two worked newform dimensions $\dim S_{12}(\Gamma_0(11))^{\mathrm{new}} = 8$ and $\dim S_{12}(\Gamma_0(2007))^{\mathrm{new}} = 1017$, all matching; (ii) confirms the sieve of Lemma 3; (iii) confirms Theorem 4 for $k = 2, \ldots, 199$; (iv) confirms the shift (1) against all 41 published terms $n = 3, \ldots, 43$, and independently against the two companion entries; and (v) confirms Luschny's formula against all 41 published terms, and against $\dim S_{n-1}(\Gamma_1(64))^{\mathrm{new}}$ for $n = 3, \ldots, 399$, with no discrepancy.

References

[1] W. A. Stein, *Modular Forms: A Computational Approach*, Graduate Studies in Mathematics 79, AMS, 2007; Chapter 6 (Dimension Formulas).

[2] F. Diamond and J. Shurman, *A First Course in Modular Forms*, Springer GTM 228, 2005; Chapter 3.

[3] G. Shimura, *Introduction to the Arithmetic Theory of Automorphic Functions*, Princeton University Press, 1994; §2.6.

[4] T. Miyake, *Modular Forms*, Springer, 1989; §2.5.

[5] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org/A063337`, consulted 24 August 2026.